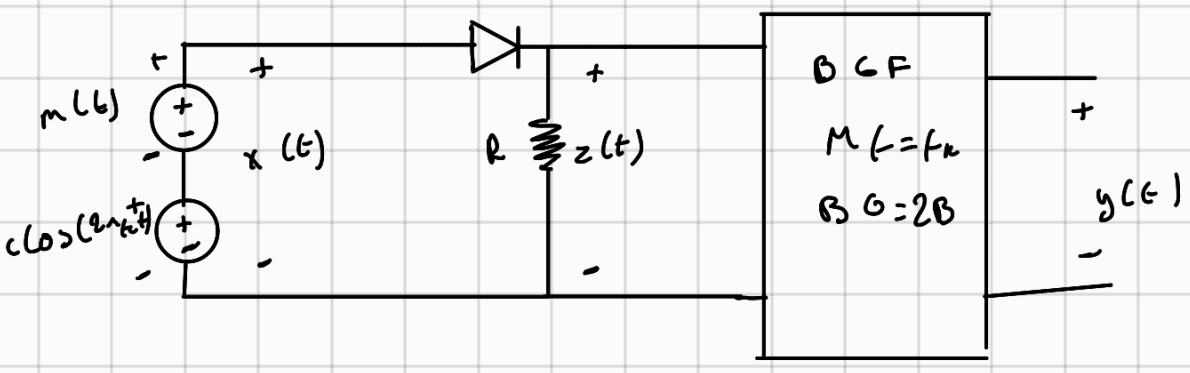
BT - GYB - GM (DSB-LC-AM)

Modülatörleri

$$y(t) = A_c[1+\mu m_n(t)]\cos(2\pi f_c t)$$

Anahtarlama modülatörü



$$x(t) = m(t) + c\cos(2\pi f_c t)$$

$$z(t) = [m(t) + c\cos(2\pi f_c t)]\,w(t)$$

$$= \left[\frac{1}{2} + \frac{2}{\pi}\left(\cos(2\pi f_c t) - \frac{1}{3}\cos(2\pi 3 f_c t) + \frac{1}{5}\cos(2\pi 5 f_c t) + \cdots\right)\right]$$

Filtreden geçmez

$$z(t) = \frac{1}{2}m(t) + \frac{2}{\pi}m(t)\cos(2\pi f_c t) + \alpha_k m(t)\cos(2\pi k f_c t)\text{'li terimler } (k=3,5,7\cdots) + \frac{c}{2}\cos(2\pi f_c t) + \frac{2c}{\pi}\cos^2(2\pi f_c t) + \beta_m \cos(2\pi m f_c t)\text{'li terimler } m=2,4,6,$$

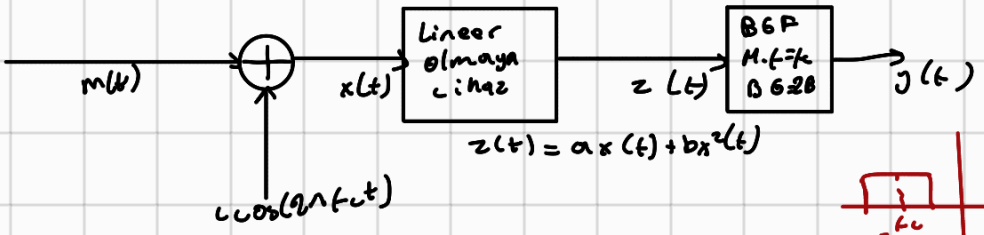
$$\cos^2(2\pi f_c t) = \frac{1}{2}[1+\cos(2\pi 2 f_c t)]$$

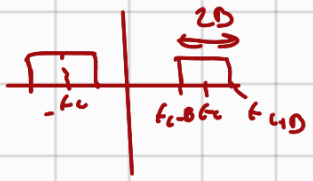
$$y(t) = \frac{c}{2}\cos(2\pi f_c t) + \frac{2}{\pi}m(t)\cos(2\pi f_c t)$$

$$= \underbrace{\frac{c}{2}}_{A_c}\left[1 + \underbrace{\frac{4}{\pi c}}_{\frac{\mu}{m_p}} m(t)\right]\cos(2\pi f_c t)$$

Lineer olmayan modülatör

(Non linear modulator. Power law modulator)





$$x(t) = m(t) + c\cos(2\pi f_c t)$$

$$z(t) = ax(t) + bx^2(t) = am(t) + ac\cos(2\pi f_c t)$$

$$+b\left[m^2(t)+c^2\cos^2(2\pi f_c t)+2m(t)c\cos(2\pi f_c t)\right]$$

$$\cos^2(2\pi f_c t)=\frac{1}{2}\left[1+\cos(2\pi 2f_c t)\right]$$

$$=am(t)+ac\cos(2\pi f_c t)+bm^2(t)+\frac{bc^2}{2}+\frac{bc^2}{2}\cos(2\pi 2f_c t)$$

$$+2bc\,m(t)\cos(2\pi f_c t)$$

$$y(t)=ac\cos(2\pi f_c t)+2bc\,m(t)\cos(2\pi f_c t)$$

$$=ac\left[1+\frac{2b}{a}m(t)\right]\cos(2\pi f_c t)$$

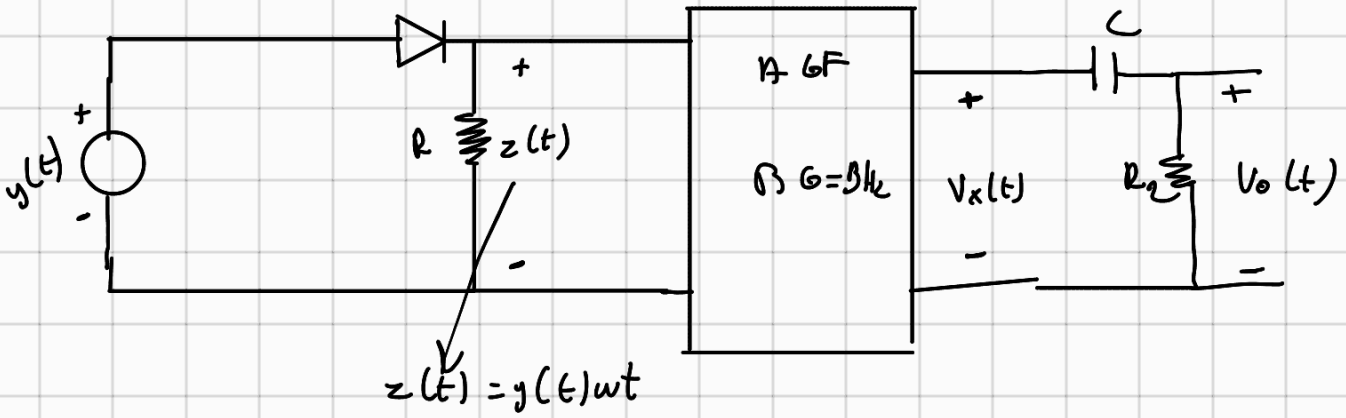
BT - GYB - GM (DSB-LC-AM)

Demodülatörler:

$$y(t)=A_c\left[1+\mu m_n(t)\right]\cos(2\pi f_c t)$$

Noncoherent demodulation (Fazı tanımsız demodülasyon)

Doğrultucu Detektör (Rectifier detector)



$$z(t)=y(t)\,w_t$$

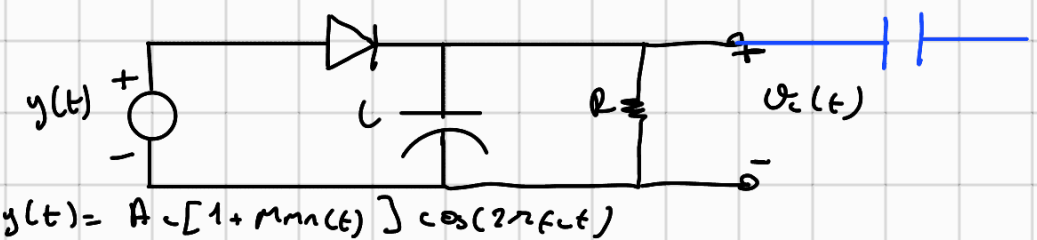
$$v_x(t)=\frac{A_c}{\pi}\left[1+\mu m_n(t)\right]$$

$$v_o(t)=\frac{A_c\mu}{\pi}m_n(t)$$

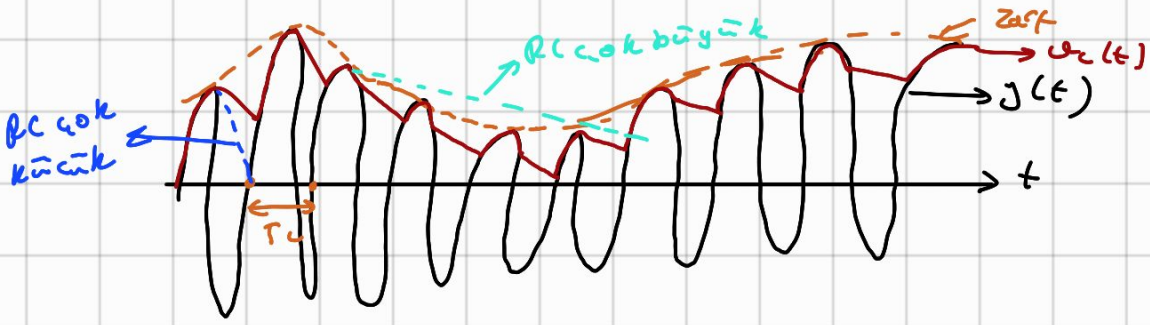
Zarf Detektörü

(Envelope detector)

$$BG\{m(t)\}=BHz \quad , f_c >> B$$



$$y(t)=A_c\left[1+\mu m_n(t)\right]\cos(2\pi f_c t)$$



$$\frac{1}{f_c} \ll RC \ll \frac{1}{B}$$

B: mesajın zamandaki değişim hızı

$$v_2(t) = a\left(\frac{A_c}{\pi}\left[1+\mu m_1(t)\right]\right) + \text{yüksek frekanslı dalgacık (ripple)}$$

$$= \alpha_1 + \alpha_2 m(t) + \text{Ripple}$$

$$\longrightarrow v_2(t) = \alpha_2 m(t) + \text{Ripple}$$

→ bundan kurtulmak için alçak geçiren geçirebilirsin

$R_L \cong 40k\Omega$

$BG\{y(t)\} = 10kHz$

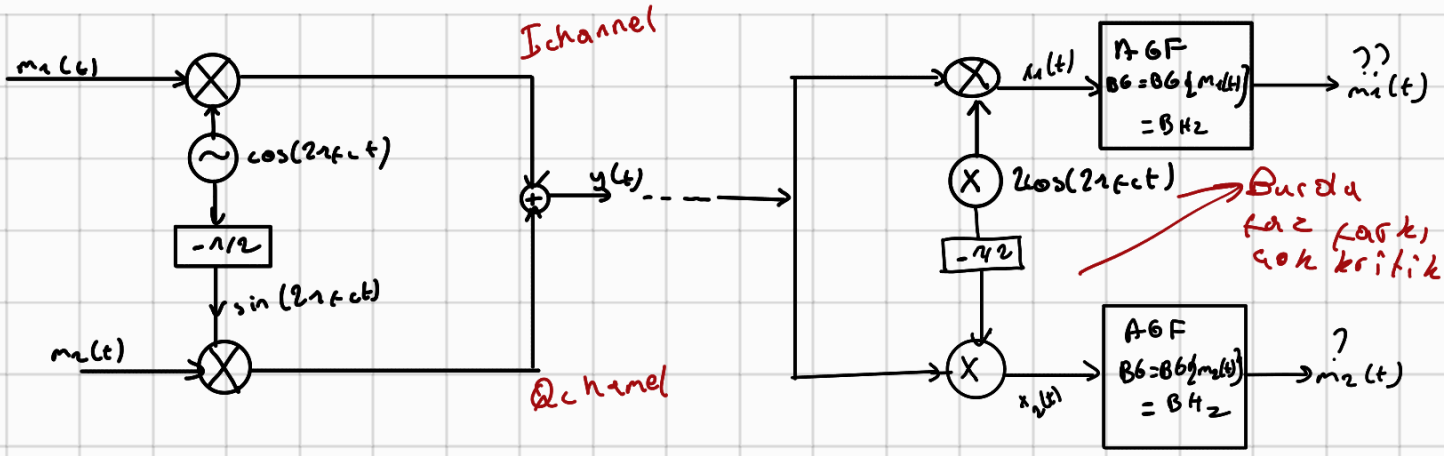
Bant genişliğini verimli kullanmak için

Quadrature Amplitude Modulation (QAM)
Dörtlü Genlik modülasyonu

Quadrature multiplexing (Gönderme)

$BG\{m_1(t)\} = BG\{m_2(t)\} = BH_z$

$$y(t) = m_1(t)\cos(2\pi f_c t) + m_2(t)\sin(2\pi f_c t)$$



$$x_1(t) = y(t)\, 2\cos(2\pi f_c t)$$

$$= 2\left[m_1(t)\cos(2\pi f_c t) + m_2(t)\sin(2\pi f_c t)\right]\cos(2\pi f_c t)$$

$$\cos a \cdot \cos b = \frac{1}{2}\left[\cos(a-b)+\cos(a+b)\right]$$

$$\sin a \cdot \sin b = \frac{1}{2}\left[\cos(a-b)-\cos(a+b)\right]$$

$$\sin a \cos b = \frac{1}{2}\left[\sin(a-b)+\sin(a+b)\right]$$

$$x(t) = 2m_1(t)\underbrace{\cos^2(2\pi f_c t)}_{\frac{1}{2}\left[1+\cos(2\pi 2f_c t)\right]} + m_2(t)\left[0+\sin(2\pi 2f_c t)\right]$$

$$x_1(t) = m_1(t) + m_1(t)\cos(2\pi 2f_c t) + m_2(t)\sin(2\pi 2f_c t)$$

geçmez geçmez

$$z_1(t) = m_1(t)$$

$$x_2(t) = y(t)\cdot 2\sin(2\pi f_c t)$$

$$= \left[m_1(t)\cos(2\pi f_c t) + m_2(t)\sin(2\pi f_c t)\right] 2\sin(2\pi f_c t)$$

$$= m_1(t)\left[0+\sin(2\pi 2f_c t)\right] + m_2(t)\left[\cos 0 - \cos(2\pi 2f_c t)\right]$$

$$= m_2(t) + m_1(t)\sin(2\pi 2f_c t) - m_2(t)\cos(2\pi 2f_c t)$$

0 0

$$z_2(t) = m_2(t)$$

✱ $BG\{m_1(t)\} = BH_z$

$BG\{m_2(t)\} = BH_z$

$4B \rightarrow BG = 2BH_z$

$$\frac{2BH_z}{2 \text{ mesaj}} = BH_z/\text{mesaj}$$

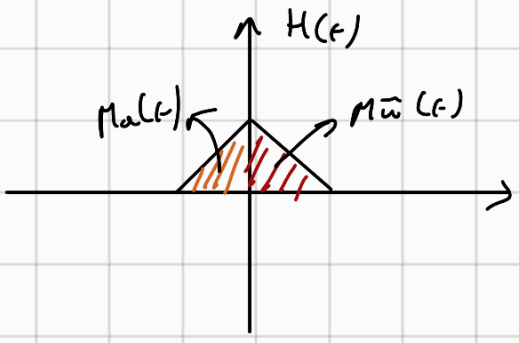
cochannel

OFDM tekniği buradan geliyor

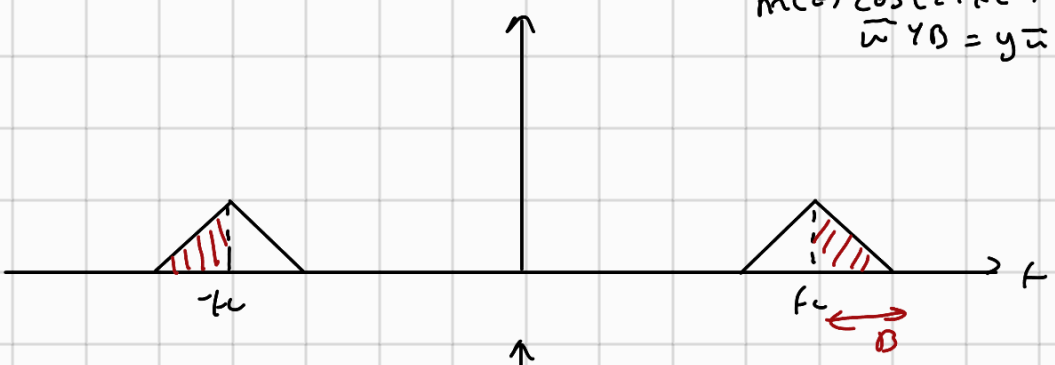
orthogonal frequency tone modulation

Tekyönlü Bandlı (TYB) GM

Single Sideband (SSB) AM



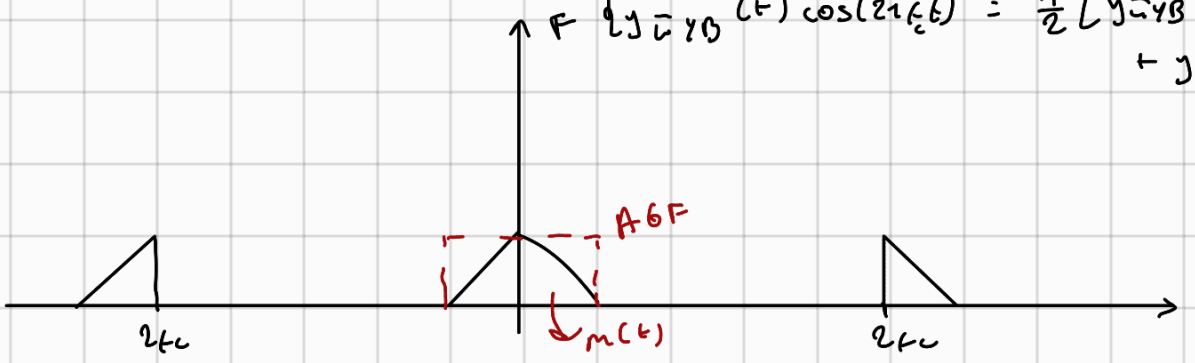
$m(t)\cos(2\pi f_c t)$ nin
üTYB = $y_{üTYB}(t)$



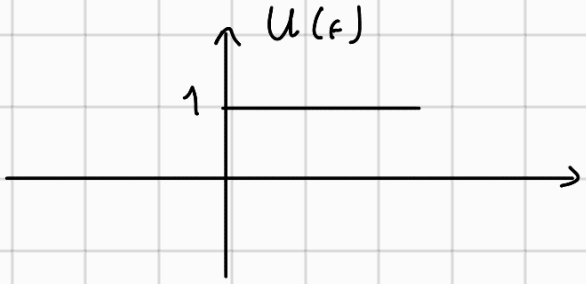
$m(t)\cos(2\pi f_c t)$ nin
ATYB = $y_{ATYB}(t)$



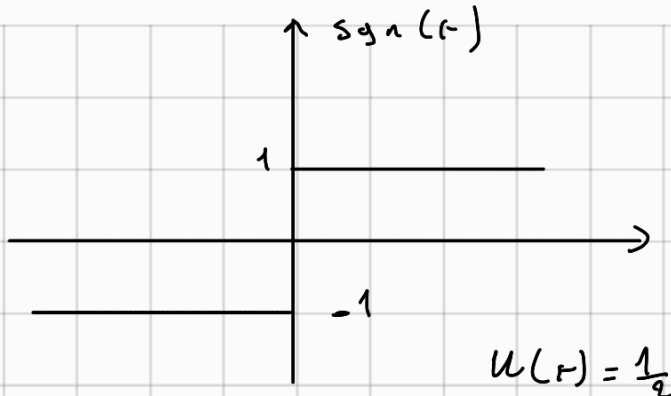
$F\{y_{üTYB}(t)\cos(2\pi f_c t)\} = \frac{1}{2}[y_{üTYB}(f-f_c) + y_{üTYB}(f+f_c)]$



$$M_{ü}(f) = M(f)\,u(f)$$



$$M_a(f) = M(f)\,u(-f)$$



$$u(f) = \frac{1}{2}\left[1 + sgn(f)\right]$$

$$M_+(f) = M(f)\,u(f)$$

$$= M(f)\,\frac{1}{2}\left[1 + sgn(f)\right]$$

$$M_+(f) = \frac{1}{2}M(f) + \frac{1}{2}M(f)\,sgn(f)$$

$\downarrow F^{-1}$

$$m_+(t) = \frac{1}{2}m(t) + \frac{1}{2}F^{-1}\{M(f)\,sgn(f)\}$$

$$F^{-1}\{M(f)\,sgn(f)\} = m(t) \circledast F^{-1}\{sgn(f)\}$$

$$x(t) = sgn(t) \xleftrightarrow{F} \frac{1}{j\pi t} = X(f)$$

$$X(t) = \frac{1}{j\pi t} \xleftrightarrow{F} sgn(-f) = x(-f)$$

$$= -sgn(f)$$

$$\frac{-1}{j\pi t} \xleftrightarrow{F} sgn(f)$$

$$F^{-1}\{sgn(f)\} = \frac{-1}{j\pi t} = \frac{j}{\pi t}$$

$$m_+(t) = \frac{1}{2}\left[m(t) + m(t) \circledast \frac{j}{\pi t}\right]$$

$$= \frac{1}{2}\left[m(t) + j\underbrace{\left(m(t) \circledast \frac{1}{\pi t}\right)}_{=m_n(t)}\right]$$

$m(t)$'nin Hilbert dönüşümü

$$m_+(t) = \frac{1}{2}\left[m(t) + j\,m_n(t)\right]$$

$$M_-(f) = M(f)\,u(-f)$$

$$= M(f)\,\frac{1}{2}\left[1 - sgn(f)\right]$$

$$= \frac{1}{2}\left[M(f) - M(f)\,sgn(f)\right]$$

$\downarrow F^{-1}$

$$m_a(t) = \frac{1}{2}[m(t) - j m_h(t)]$$

$$Y_{\tilde{u}_{YB}}(t) = M_{\tilde{u}}(f - f_c) + M_a(f + f_c)$$

$\downarrow F^{-1}$

$$y_{\tilde{u}_{YB}}(t) = m_{\tilde{u}}(t) e^{j2\pi f_c t} + m_a(t) e^{-j2\pi f_c t}$$

$$= \frac{1}{2}[m(t) + j m_h(t)] e^{j2\pi f_c t} + \frac{1}{2}[m(t) - j m_h(t)] e^{-j2\pi f_c t}$$

$$y_{\tilde{u}_{YB}}(t) = m(t)\cos(2\pi f_c t) + m_h(t)\left[\frac{j e^{j2\pi f_c t} - j e^{-j2\pi f_c t}}{2}\right]$$

$$j \cdot j = -1 \qquad \left[\frac{e^{j2\pi f_c t} - e^{-j2\pi f_c t}}{2j}\right] = \sin(2\pi f_c t)$$

$$y_{\tilde{u}_{YB}}(t) = m(t)\cos(2\pi f_c t) - m_h(t)\sin(2\pi f_c t)$$

$$y_{\tilde{u}_{YB}}(t) = m(t)\cos(2\pi f_c t) - m_h(t)\sin(2\pi f_c t)$$

$$Y_{A_{YB}}(t) = M_a(f - f_c) + M_{\tilde{u}}(f + f_c)$$

$$y_{A_{YB}}(t) = m_a(t) e^{j2\pi f_c t} + m_{\tilde{u}}(t) e^{-j2\pi f_c t}$$

$m_a(t) \rightarrow \frac{1}{2}[m(t) - j m_h(t)]$ $\qquad$ $m_{\tilde{u}}(t) \rightarrow \frac{1}{2}[m(t) + j m_h(t)]$

$$y_{\tilde{u}_{YB}}(t) = m(t)\cos(2\pi f_c t) \mp m_h(t)\sin(2\pi f_c t)$$

$\tilde{u}_{YB}$: $-$ $\qquad$ A_{YB} : $+$

hilbert dönüşümü mesaj işaretine göre